

Introduction to Data Science  
Course Project  
Report Document

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21L-5692

BDS-3A

## **Instructions: Read These Carefully Before Starting**

1. Due Date: Sunday 4th December 2022 – 11:59PM
2. Submission will be taken on Google Classroom
3. Submit only the following 2 files named like the following:
  - a. Code File (Jupyter Notebook): L210000\_Code.ipynb
  - b. Report Document (This File): L210000\_Report.pdf
4. Project will not be evaluated if:
  - a. You submit python (.py) files
  - b. You submit multiple .ipynb files
  - c. You submit compressed (.rar or .zip) files
  - d. You submit any files other than the required PDF and IPYNB
5. Upload data files directly to Google Colab - do not use Google Drive or GitHub linking method
6. All source files needed to complete this project are uploaded with it on Google Classroom.
7. Do not add the data file with your submission on Google Classroom.

Not following these instructions will lead to mark deduction.

**Please try to use Microsoft Word instead of Google Docs to edit this document and to export it as a PDF file for final submission.**

Happy Coding 🐱

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*TA Emails*

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For this project you will be applying machine learning models (both regression and classification) to the dataset which contains information about various individuals, their clothing, and its properties along with other atmospheric elements such as temperature, pressure humidity etc. The users also provided feedback on if they feel cold or not. The feedback (through AMV and PMV) which is based on the following mapping:

The following table shows the mapping of sensations:

| Value | Thermal Sensation |
|-------|-------------------|
| +3    | hot               |
| +2    | warm              |
| +1    | slightly warm     |
| 0     | neutral           |
| -1    | slightly cool     |
| -2    | cool              |
| -3    | cold              |

**The dataset is given in an excel file named CollectedData.xlsx, see sheet 2 of excel file.** The dimension names (column headers) are not mentioned in the given file. The table below describes the columns which will be of your interest.

| Column number | Feature Name | Feature Description              |
|---------------|--------------|----------------------------------|
| 3             | Age          | Age                              |
| 22            | Clo          | Clothing insulation              |
| 19            | Met          | Met Rate                         |
| 26            | Dewpt        | Dewpt                            |
| 27            | PlaneRadTemp | plane radiant temperature        |
| 37            | Ta           | Average air temperature          |
| 38            | Tmrt         | Average mean radiant temperature |
| 40            | Vel          | Air Velocity                     |
| 42            | AirTurb      | Air Turbulance                   |
| 43            | Pa           | Vapor Pressure                   |
| 44            | Rh           | Humidity                         |
| 74            | TaOutdoor    | Outdoor Air Temperature          |
| 77            | RhOutdoor    | Outdoor Humidity                 |
| 8             | AMV          | Classification response variable |
| 49            | PMV          | Regression response variable     |

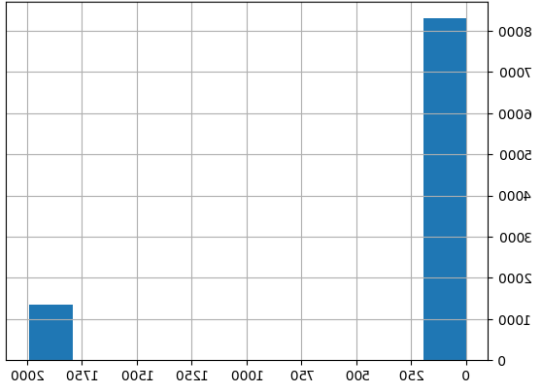
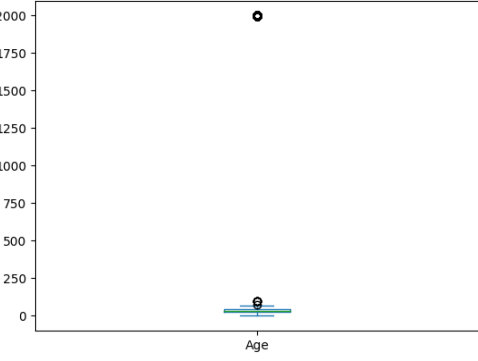
### **Part A. Preprocessing**

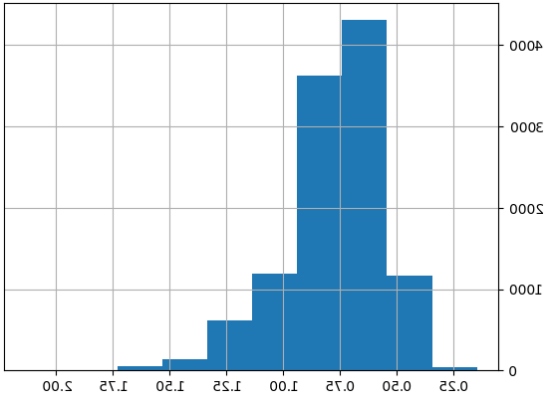
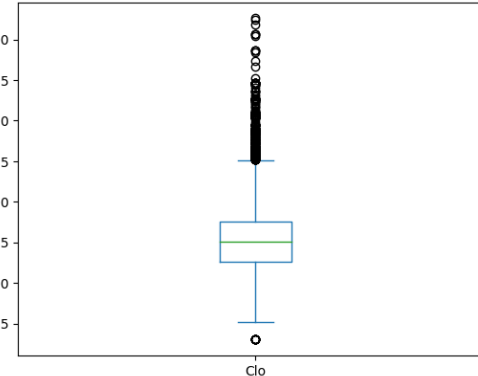
**1. In this step, you are required to apply the preprocessing steps that you've covered in the course. Specifically, for each of the input dimension, fill in the following (add rows and complete the table for all input dimensions).**

| Dim Name     | Data Type | Total Instances | Number of Nulls | Number of Outliers | Min. Value | Max Value | Mode | Mean       | Median    | Variance     | STD        |
|--------------|-----------|-----------------|-----------------|--------------------|------------|-----------|------|------------|-----------|--------------|------------|
| Age          | float64   | 9650            | 2915            | 1349               | 0.000      | 1996.0000 | 24.0 | 308.637202 | 35.000000 | 462556.56104 | 680.115105 |
| Clo          | float64   | 11159           | 1406            | 782                | 0.150      | 2.13000   | 0.77 | 0.778503   | 0.751700  | 0.049284     | 0.221999   |
| Met          | float64   | 10678           | 1887            | 2127               | 0.100      | 4.50000   | 1.0  | 1.065907   | 1.100000  | 0.183939     | 0.428882   |
| Dewpt        | float64   | 9014            | 3551            | 789                | -1.953     | 26.89675  | 17.4 | 13.621447  | 14.100000 | 34.845928    | 5.903044   |
| PlaneRadTemp | float64   | 5544            | 7021            | 5544               | -7.420     | 11.70000  | 0.3  | 0.217785   | 0.200000  | 1.084022     | 1.041164   |
| Ta           | float64   | 12545           | 20              | 539                | 15.960     | 31.00000  | 23.2 | 23.179187  | 23.136667 | 2.053443     | 1.432984   |
| Tmrt         | float64   | 8864            | 3701            | 1620               | 16.610     | 37.44500  | 22.5 | 23.450693  | 23.359219 | 2.257475     | 1.502489   |
| Vel          | float64   | 8865            | 3700            | 1261               | 0.000      | 1.88000   | 0.1  | 0.112445   | 0.100000  | 0.006248     | 0.079044   |
| AirTurb      | float64   | 6965            | 5600            | 1444               | 0.000      | 102.45000 | 0.5  | 18.265870  | 0.500000  | 627.057129   | 25.04109   |
| Pa           | float64   | 7910            | 4655            | 1349               | 0.000      | 27.70000  | 2.1  | 5.123996   | 1.550667  | 66.522562    | 8.156136   |

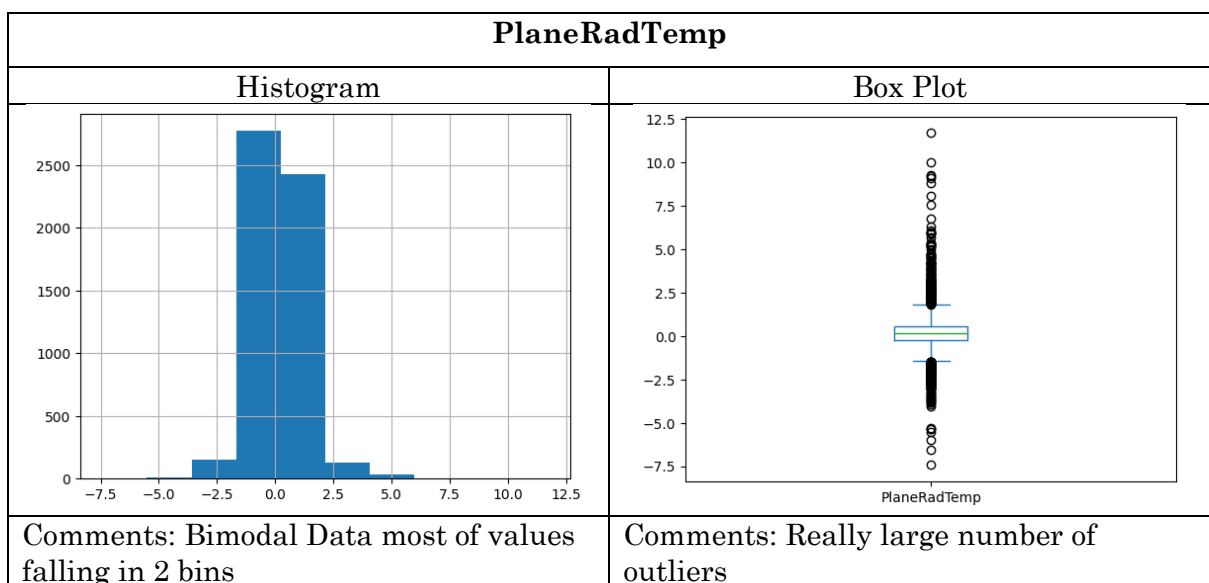
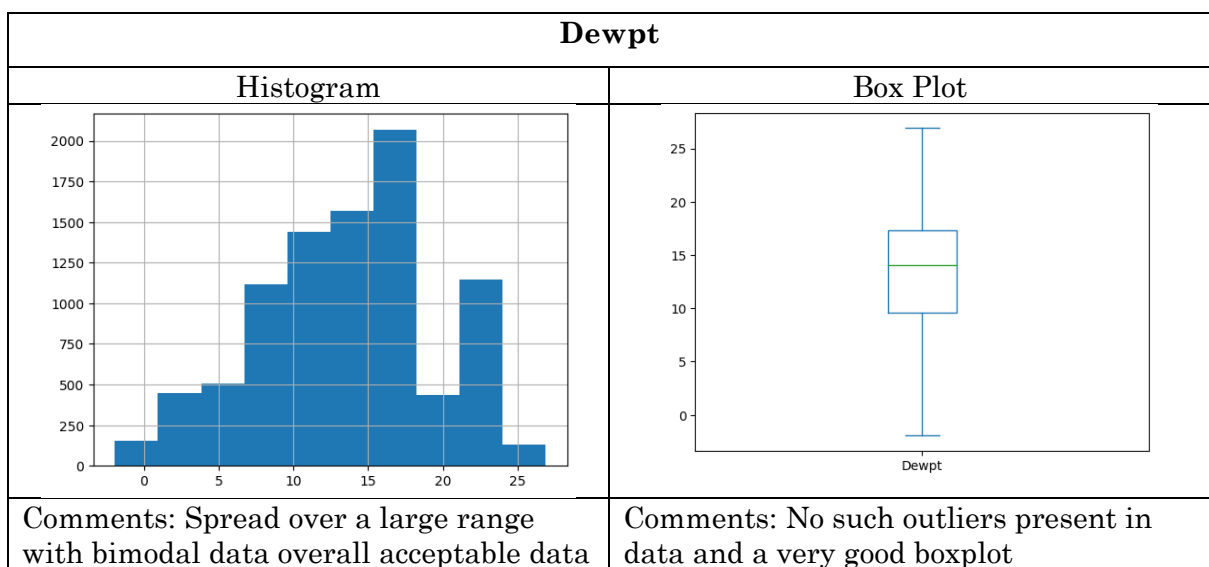
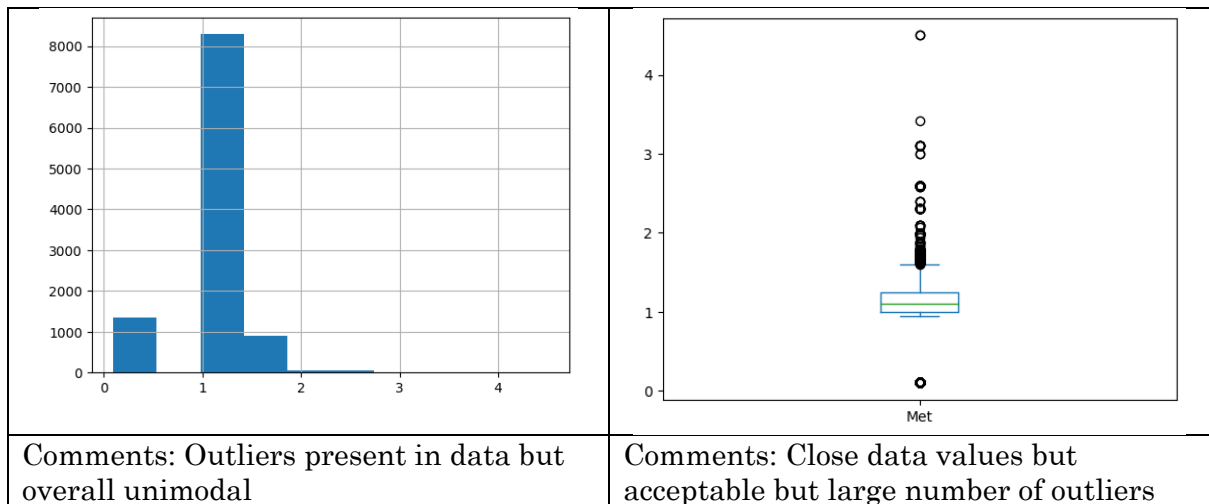
|           |             |           |      |      |                 |                   |               |                           |                   |                    |                           |
|-----------|-------------|-----------|------|------|-----------------|-------------------|---------------|---------------------------|-------------------|--------------------|---------------------------|
| Rh        | float6<br>4 | 1253<br>0 | 35   | 0    | 7.4<br>00       | 79.<br>300<br>00  | 64.0          | 42<br>.5<br>28<br>50<br>7 | 43.<br>276<br>815 | 226.<br>8480<br>17 | 15<br>.0<br>61<br>47<br>5 |
| TaOutdoor | float6<br>4 | 1119<br>7 | 1368 | 221  | -24<br>.90<br>0 | 32.<br>350<br>00  | 27.5555<br>56 | 17<br>.1<br>75<br>08<br>7 | 18.<br>200<br>000 | 113.<br>7510<br>67 | 10<br>.6<br>65<br>41<br>5 |
| RhOutdoor | float6<br>4 | 1254<br>6 | 19   | 1349 | 0.00<br>0       | 100<br>.35<br>000 | 0.0           | 61<br>.0<br>98<br>93<br>9 | 68.<br>795<br>799 | 610.<br>3055<br>83 | 24<br>.7<br>04<br>36<br>4 |
| AMV       | float6<br>4 | 1251<br>0 | 55   | 0    | -3.<br>000      | 3.0<br>000<br>0   | 0.0           | 0.<br>10<br>05<br>84      | 0.0<br>000<br>00  | 1.21<br>4430       | 1.<br>10<br>20<br>12      |
| PMV       | float6<br>4 | 1186<br>9 | 696  | 346  | -4.<br>170      | 2.5<br>000<br>0   | 0.1           | -0<br>.0<br>73<br>71<br>1 | -0.<br>030<br>000 | 0.28<br>9471       | 0.<br>53<br>80<br>25      |

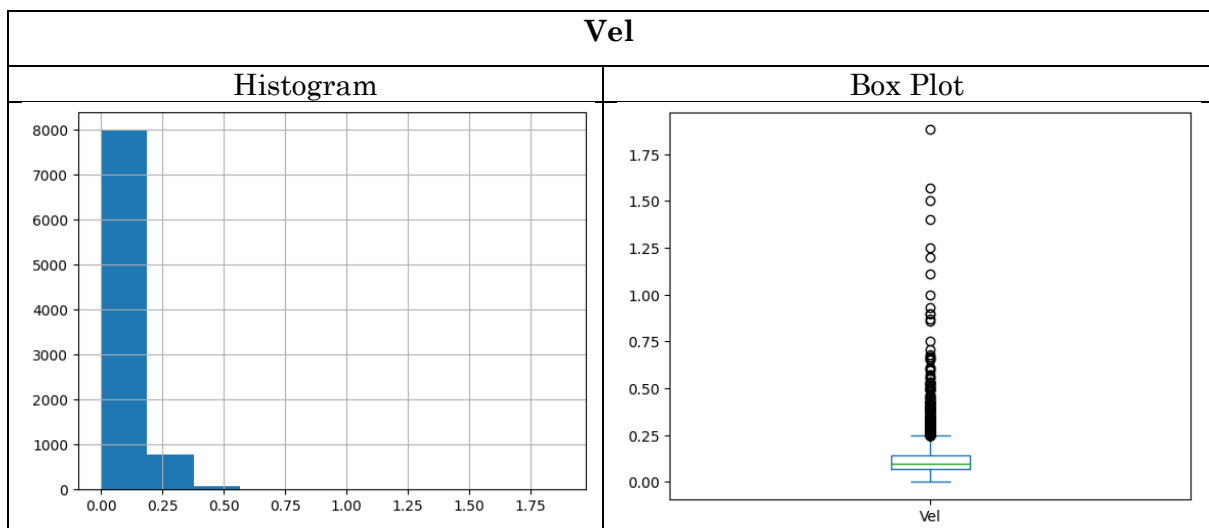
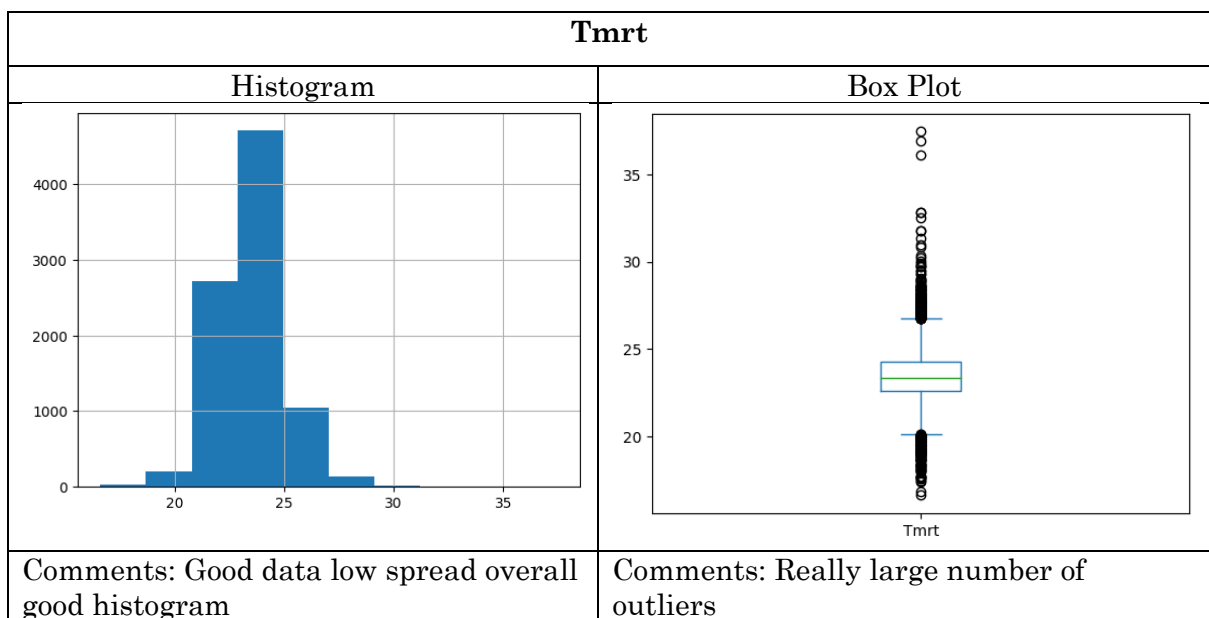
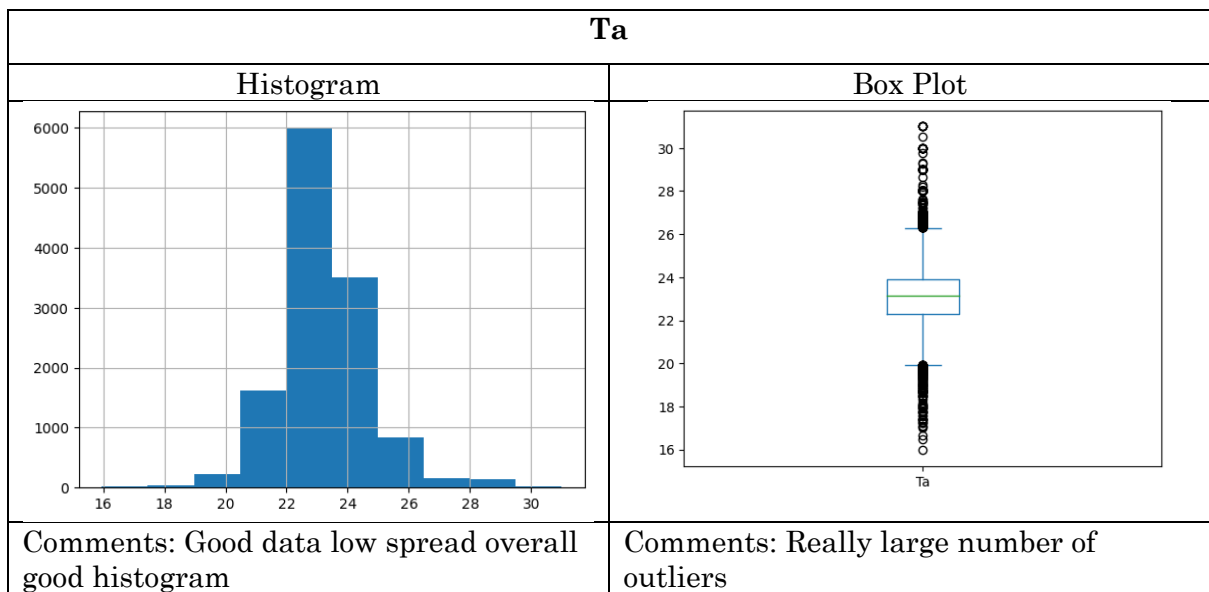
2. For each of the input dimension, plot histogram and comment the type of distribution the dimension exhibits. Further, visualize each dimension using a Box Plot. Specifically, for each of the input dimension, you're required to fill the following table (duplicate it for each of the 15 dimensions).

| Age                                                                                                         |                                                                                          |
|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Histogram                                                                                                   | Box Plot                                                                                 |
|                            |        |
| Comments: Substantial Amount of Outliers in the data and which are actual data is a good data unimodal data | Comments: Data is very close to each other and outliers are very large and very far away |

| Clo                                                                                 |                                                                                      |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Histogram                                                                           | Box Plot                                                                             |
|  |  |
| Comments: A bit Left Skewed Data with 2 good bins                                   | Comments: Large number of outliers other than that good                              |

| Met       |          |
|-----------|----------|
| Histogram | Box Plot |



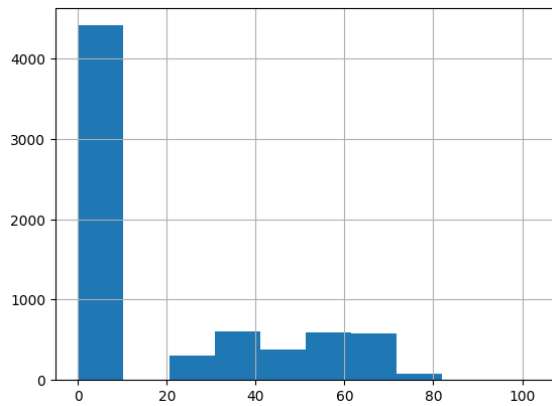


Comments: 90 percent of data falling into 1 bin and no such outlier

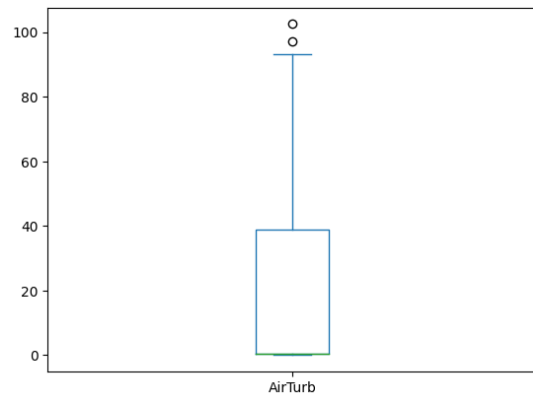
Comments: Really large number of outliers

### AirTurb

Histogram



Box Plot

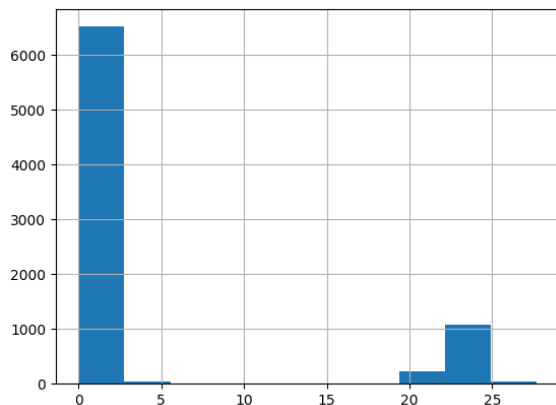


Comments: Outliers present in data with a good bin

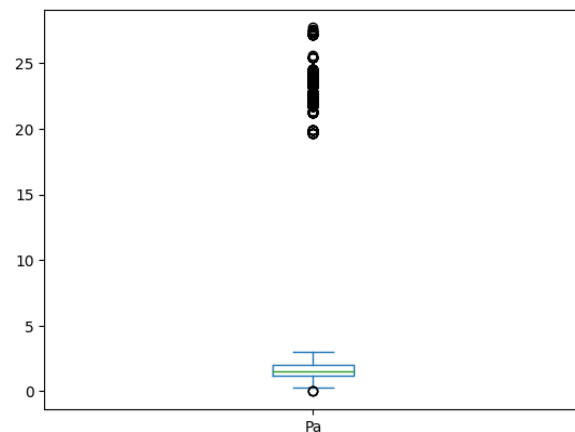
Comments: Data is very close to minimum value

### Pa

Histogram



Box Plot



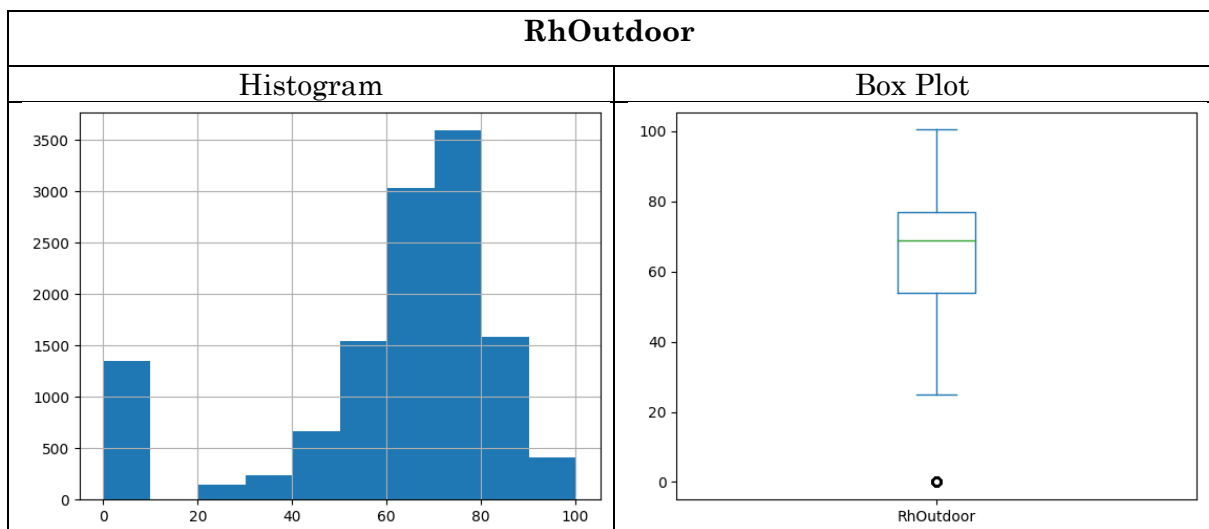
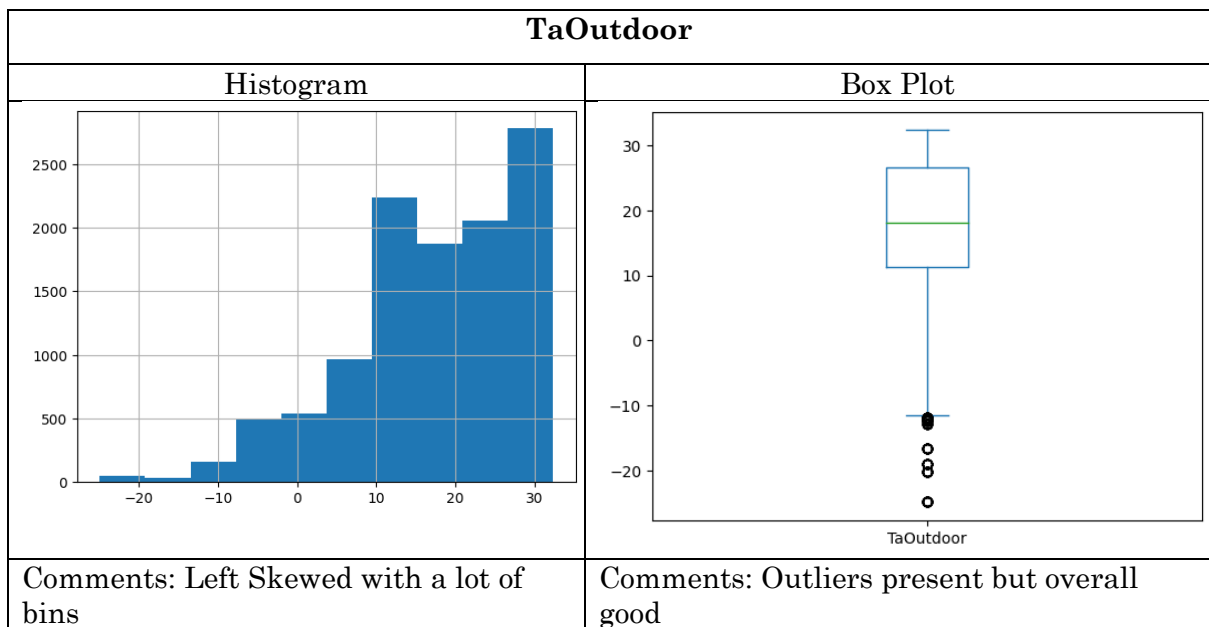
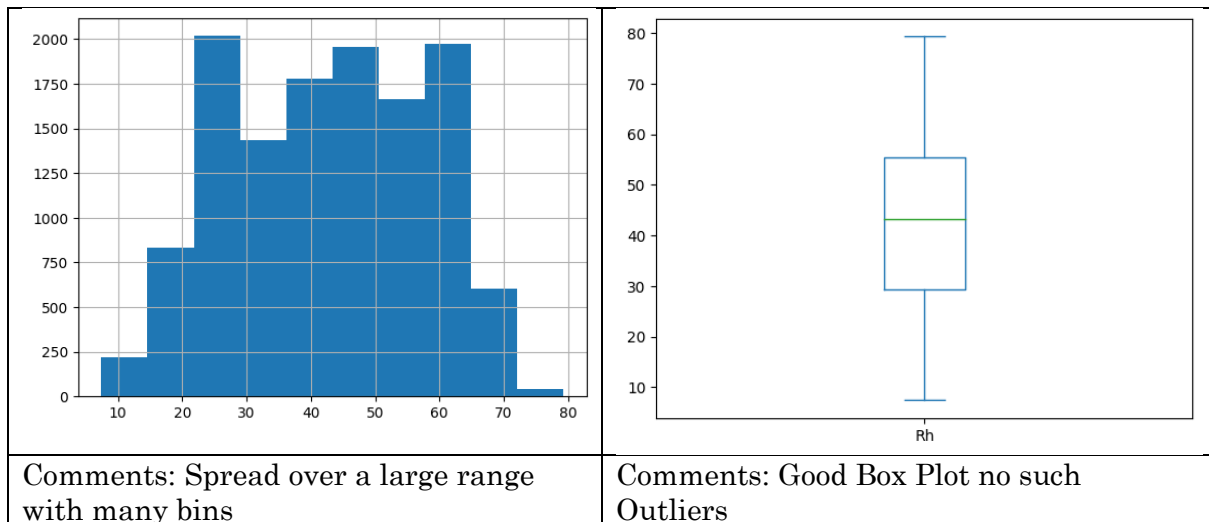
Comments: Big outliers present in data

Comments: Really large number of outliers

### Rh

Histogram

Box Plot

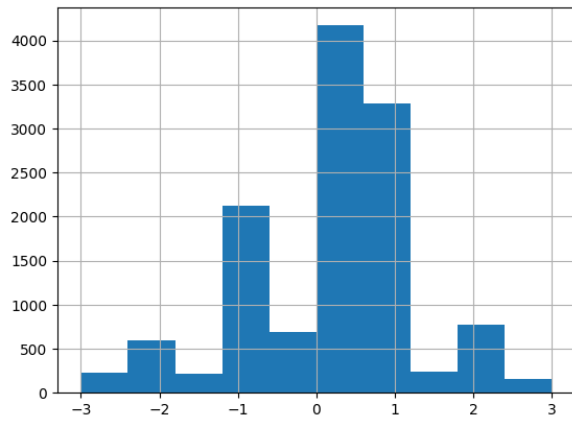


Comments: Substantial amount of outliers present in data if those omitted then good spread and values

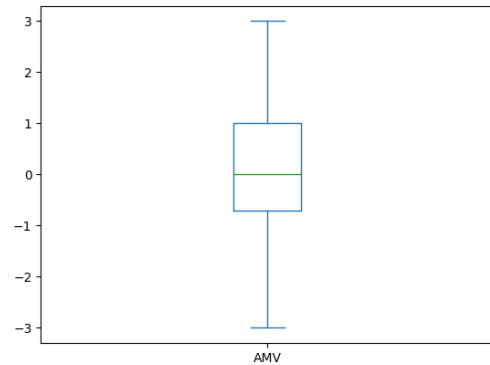
Comments: Good Box Plot

### AMV

Histogram



Box Plot

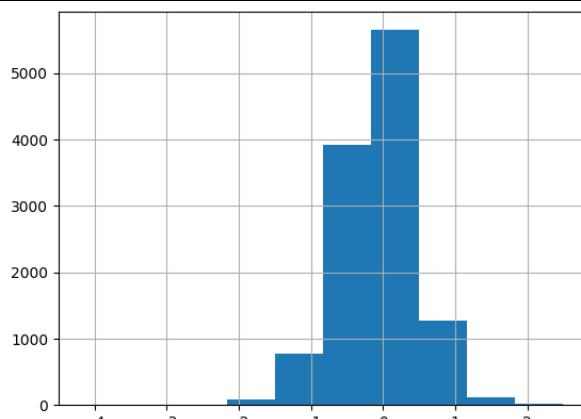


Comments: Range is spread but data dispersed over a lot of bins

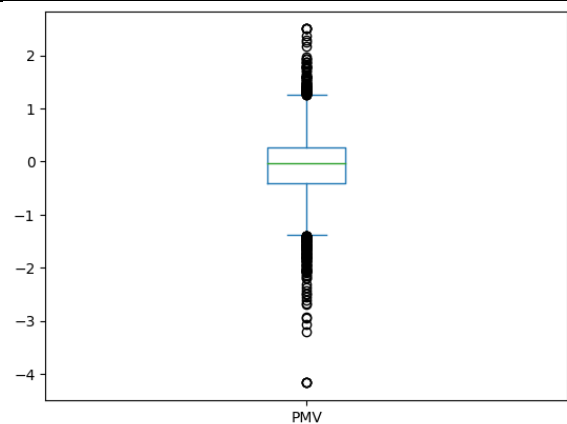
Comments: Good Box Plot

### PMV

Histogram



Box Plot



Comments: Good number of bins with good data placement a really good histogram

Comments: Really large number of outliers

**3. Find the missing values in each of the dimension (do this for both input and output dimensions), and fill these using an “appropriate” methodology that we’ve discussed in the class. You may also choose to drop a certain sample based on your analysis. Mention your approach and its justification.**

| Dim Name     | Number of Missing Values | Filled using OR Dropped | Reason for selecting a certain approach                                                                                |
|--------------|--------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------|
| Age          | 2915                     | Filled Using Mean       | Data is near to each other and spread is acceptable hence better value will come from mean rather than median and mode |
| Clo          | 1406                     | Filled Using Mean       | Data is near to each other and spread is acceptable hence better value will come from mean rather than median and mode |
| Met          | 1887                     | Filled Using Mean       | Data is near to each other and spread is acceptable hence better value will come from mean rather than median and mode |
| Dewpt        | 3551                     | Filled Using Mean       | Data is near to each other and spread is acceptable hence better value will come from mean rather than median and mode |
| PlaneRadTemp | 7021                     | Filled Using Mean       | Data is near to each other and spread is acceptable hence better value will come from mean rather than median and mode |
| Ta           | 20                       | Filled Using Mean       | Data is near to each other and spread is acceptable hence better value will come from mean rather than median and mode |

|           |      |                   |                                                                                                                        |
|-----------|------|-------------------|------------------------------------------------------------------------------------------------------------------------|
| Tmrt      | 3701 | Filled Using Mean | Data is near to each other and spread is acceptable hence better value will come from mean rather than median and mode |
| Vel       | 3700 | Filled Using Mean | Data is near to each other and spread is acceptable hence better value will come from mean rather than median and mode |
| AirTurb   | 5600 | Filled Using Mean | Data is near to each other and spread is acceptable hence better value will come from mean rather than median and mode |
| Pa        | 4655 | Filled Using Mean | Data is near to each other and spread is acceptable hence better value will come from mean rather than median and mode |
| Rh        | 35   | Filled Using Mean | Data is near to each other and spread is acceptable hence better value will come from mean rather than median and mode |
| TaOutdoor | 1368 | Filled Using Mean | Data is near to each other and spread is acceptable hence better value will come from mean rather than median and mode |
| RhOutdoor | 19   | Filled Using Mean | Data is near to each other and spread is acceptable hence better value will come from mean rather than median and mode |
| AMV       | 55   | Filled Using Mean | Data is near to each other and spread is acceptable hence                                                              |

|     |     |                   |                                                                                                                        |
|-----|-----|-------------------|------------------------------------------------------------------------------------------------------------------------|
|     |     |                   | better value will come from mean rather than median and mode                                                           |
| PMV | 696 | Filled Using Mean | Data is near to each other and spread is acceptable hence better value will come from mean rather than median and mode |

**4. For each of the dimension, find out the outliers (noisy data) and handle these appropriately.**

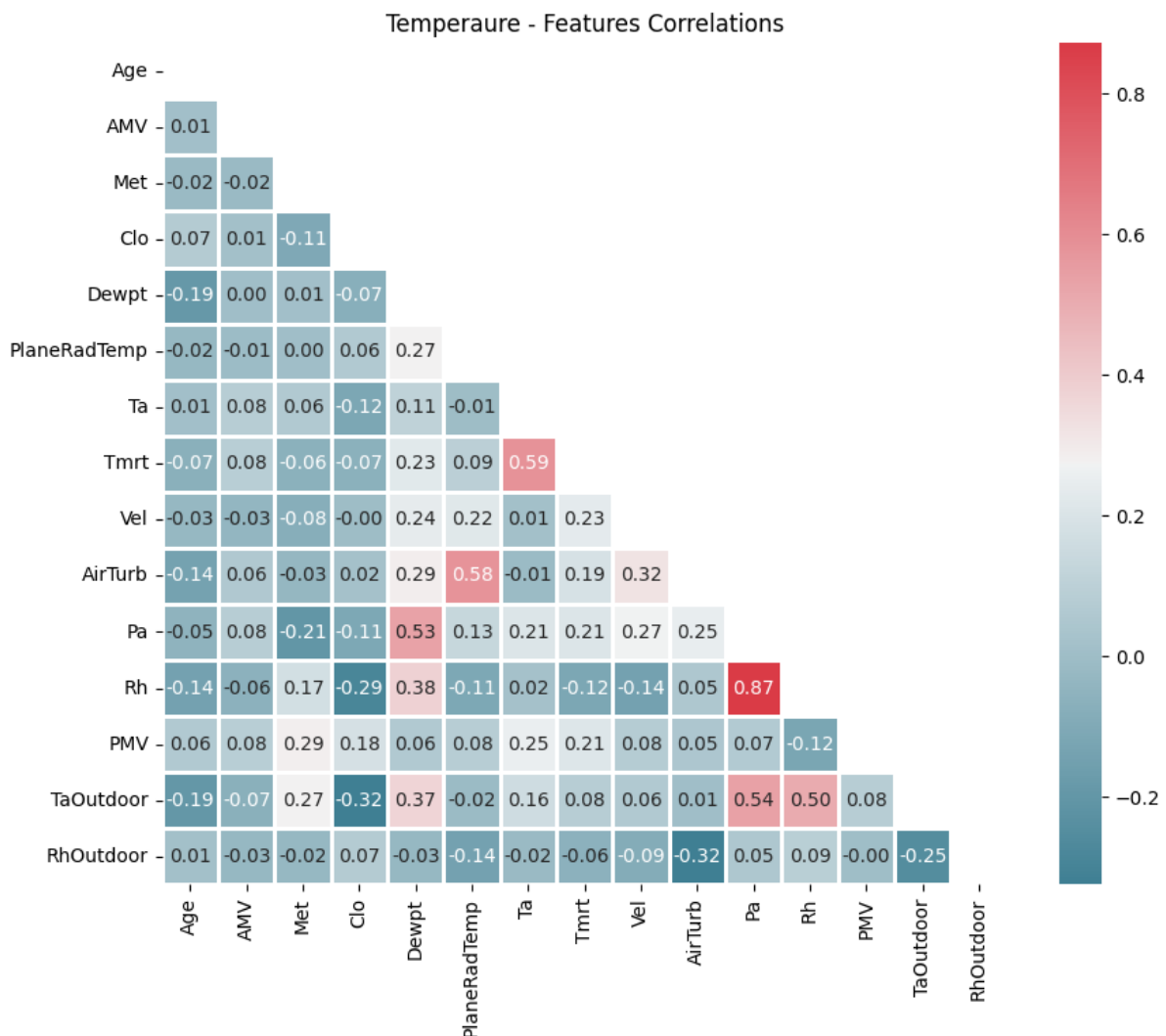
| Dim Name     | Number of Outliers | Smooth using/<br>Dropped | Reason for selecting<br>a certain approach              |
|--------------|--------------------|--------------------------|---------------------------------------------------------|
| Age          | 1349               | Smoothed Using IQR       | it tells us inside what range the bulk of our data lies |
| Clo          | 782                | Smoothed Using IQR       | it tells us inside what range the bulk of our data lies |
| Met          | 2127               | Smoothed Using IQR       | it tells us inside what range the bulk of our data lies |
| Dewpt        | 789                | Smoothed Using IQR       | it tells us inside what range the bulk of our data lies |
| PlaneRadTemp | 5544               | Smoothed Using IQR       | it tells us inside what range the bulk of our data lies |
| Ta           | 539                | Smoothed Using IQR       | it tells us inside what range the bulk of our data lies |
| Tmrt         | 1620               | Smoothed Using IQR       | it tells us inside what range the bulk of our data lies |
| Vel          | 1261               | Smoothed Using IQR       | it tells us inside what range the bulk of our data lies |
| AirTurb      | 1444               | Smoothed Using IQR       | it tells us inside what range the bulk of our data lies |
| Pa           | 1349               | Smoothed Using IQR       | it tells us inside what range the bulk of our data lies |
| Rh           | 0                  | Smoothed Using IQR       | it tells us inside what range the bulk of our data lies |
| TaOutdoor    | 221                | Smoothed Using IQR       | it tells us inside what range the bulk of our data lies |
| RhOutdoor    | 1349               | Smoothed Using IQR       | it tells us inside what range the bulk of our data lies |
| AMV          | 0                  | Smoothed Using IQR       | it tells us inside what range the bulk of our data lies |

|     |     |                    |                                                         |
|-----|-----|--------------------|---------------------------------------------------------|
| PMV | 346 | Smoothed Using IQR | it tells us inside what range the bulk of our data lies |
|-----|-----|--------------------|---------------------------------------------------------|

**5. Using the variance that you've calculated above, for each dimension, comment whether you'll select the input dimension or no. (don't drop a dimension at this point)**

| Dim Name     | Variance      | Apply filter or no, reason |
|--------------|---------------|----------------------------|
| Age          | 462556.556104 | NO                         |
| Clo          | 0.049284      | NO                         |
| Met          | 0.183939      | NO                         |
| Dewpt        | 34.845928     | NO                         |
| PlaneRadTemp | 1.084022      | NO                         |
| Ta           | 2.053443      | NO                         |
| Tmrt         | 2.257475      | NO                         |
| Vel          | 0.006248      | Yes , Variance is too low  |
| AirTurb      | 627.057129    | NO                         |
| Pa           | 66.522562     | NO                         |
| Rh           | 226.848017    | NO                         |
| TaOutdoor    | 113.751067    | NO                         |
| RhOutdoor    | 610.305583    | NO                         |
| AMV          | 1.214430      | NO                         |
| PMV          | 0.289471      | NO                         |

**6A. Create a correlation matrix (Heat Map) for all the dimensions (input and output).**



**6B. Using the above correlation matrix, comment what are the most informative dimensions, and which are the least. Note that, be careful since we have two response variables in the dataset (i.e., PMV and AMV regression and classification respectively)**

RhOutdoor , Vel , Ta , PhonRadTemp and Met are the least informative dimensions whereas Dewpt , AirTurb , Rh and TaOutdoor are the most informative dimensions due to their high co-relation with the response variables.

**7. Apply entropy followed by information gain on the selected columns. Specify your selection criteria.**

| Dim name     | Entropy | Info Gain           | Reason                                                                                    |
|--------------|---------|---------------------|-------------------------------------------------------------------------------------------|
| Age          | 4.519   | 3.5950000000000006  | As data is spread and to remove any impurity present in data I have applied entropy on it |
| Clo          | 6.683   | 1.4310000000000001  | As data is spread and to remove any impurity present in data I have applied entropy on it |
| Met          | 4.39    | 3.7240000000000001  | As data is spread and to remove any impurity present in data I have applied entropy on it |
| Dewpt        | 6.983   | 1.1310000000000001  | As data is spread and to remove any impurity present in data I have applied entropy on it |
| PlaneRadTemp | 3.404   | 4.7100000000000001  | As data is spread and to remove any impurity present in data I have applied entropy on it |
| Ta           | 7.713   | 0.4010000000000007  | As data is spread and to remove any impurity present in data I have applied entropy on it |
| Tmrt         | 6.922   | 1.1920000000000001  | As data is spread and to remove any impurity present in data I have applied entropy on it |
| AirTurb      | 4.928   | 3.1860000000000001  | As data is spread and to remove any impurity present in data I have applied entropy on it |
| Pa           | 10.256  | -2.1419999999999995 | As data is spread and to remove any impurity present in data I have applied entropy on it |
| Rh           | 10.702  | -2.5879999999999999 | As data is spread and to remove any impurity present in                                   |

|           |       |                    |                                                                                           |
|-----------|-------|--------------------|-------------------------------------------------------------------------------------------|
|           |       |                    | data I have applied entropy on it                                                         |
| TaOutdoor | 7.301 | 0.8130000000000006 | As data is spread and to remove any impurity present in data I have applied entropy on it |
| RhOutdoor | 7.307 | 0.8070000000000004 | As data is spread and to remove any impurity present in data I have applied entropy on it |
| AMV       | 2.707 | 5.4070000000000001 | As data is spread and to remove any impurity present in data I have applied entropy on it |
| PMV       | 6.8   | 1.3140000000000001 | As data is spread and to remove any impurity present in data I have applied entropy on it |

### **Part B. Applying Algorithms**

1. For this part, split the data randomly into 80/20 percent. Where 80% represents the training data. Also normalize the dataset as you see fit.

**2A. Apply forward selection, considering PMV as response variable and Multilinear regression as machine learning model.** Create a table, that mentions dimensions, and performance achieved. Which is the optimal feature set, and why.

| Feature Vector | Performance achieved |
|----------------|----------------------|
| Met            | Yes                  |
| Clo            | Yes                  |
| Ta             | Yes                  |
| Tmrt           | Yes                  |
| Rh             | Yes                  |

This is the optimal feature set with 5 features that are giving the maximum performance in the multilinear regression machine algorithm and our classifier is performing best with these 5 features when we are applying forward selection

**2B. Apply backward selection, considering PMV as response variable and Multilinear regression as machine learning model.** Create a table, that mentions dimensions, and performance achieved. Which is the optimal feature set, and why.

| Feature Vector | Performance achieved |
|----------------|----------------------|
| Met            | Yes                  |
| Clo            | Yes                  |
| Ta             | Yes                  |
| Pa             | Yes                  |
| Rh             | Yes                  |

This is the optimal feature set with 5 features that are giving the maximum performance in the multilinear regression machine algorithm and our classifier is performing best with these 5 features when we are applying backward selection

**3A. Apply forward selection, considering AMV as response variable and Logistic regression as machine learning model.** Create a table, that mentions dimensions, and performance achieved. Which is the optimal feature set, and why.

| Feature Vector | Performance achieved |
|----------------|----------------------|
| Met            | Yes                  |
| Clo            | Yes                  |
| Ta             | Yes                  |
| Tmrt           | Yes                  |
| Rh             | Yes                  |

This is the optimal feature set with 5 features that are giving the maximum performance in the logistic regression machine algorithm and our classifier is performing best with these 5 features when we are applying forward selection

**3B. Apply backward selection, considering AMV as response variable and Logistic regression as machine learning model.** Create a table, that mentions dimensions, and performance achieved. Which is the optimal feature set, and why.

| Feature Vector | Performance achieved |
|----------------|----------------------|
| Age            | Yes                  |
| Met            | Yes                  |
| Clo            | Yes                  |
| Dewpt          | Yes                  |
| PMV            | Yes                  |

This is the optimal feature set with 5 features that are giving the maximum performance in the logistic regression machine algorithm and our classifier is performing best with these 5 features when we are applying backward selection

4. Using the optimal feature vector that you've figured out from your analysis above, apply 3-fold cross validation for both regression and classification problems (PMV and AMV respectively). Write down the optimal parameters values for each of the model. Further, plot confusion matrix for the classification part.

| Optimal Parameters     |                     |
|------------------------|---------------------|
| Multilinear Regression | Logistic Regression |
| Met                    | Age                 |
| Clo                    | Met                 |
| Ta                     | Clo                 |
| Tmrt                   | Dewpt               |
| Rh                     | PMV                 |

